

Economic Weather Industry Rotation

MGMTMFE 431 Quantitative Asset Management Final Project

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Introduction and Motivation

This project studies whether an industry-rotation strategy can improve the risk-adjusted performance of a fully invested equity portfolio. Industries are treated as economic habitats that respond differently to macro “weather.” This weather overlay is combined with factor forecasts and industry momentum, then passed through a constrained active optimizer.

The main asset universe is the Ken French 49 Industry Portfolios. After aligning the industry returns with the Fama-French factor data, AQR factor data, and FRED macroeconomic data, the backtest uses 47 monthly industry excess return series. The benchmark portfolios are the equal-weighted industry portfolio and the market excess return from the Fama-French data. As a robustness check, the same construction is also applied to the Ken French 10 Industry Portfolios.

Portfolio Construction

The strategy is rebalanced monthly. At each month-end, it uses the prior 120 months to estimate an augmented eight-factor model: the Fama-French five factors, Ken French momentum, AQR Quality Minus Junk, and AQR Betting Against Beta. For industry i ,

$$r_{i,\tau}^e = \alpha_i + \beta_i' f_\tau + \epsilon_{i,\tau}.$$

The expected excess return forecast combines four pieces:

$$\hat{\mu}_{i,t} = \tilde{\alpha}_{i,t} + \hat{\beta}_{i,t}' \tilde{f}_t + 0.012z(\text{MOM}_{i,t}) + \omega_{i,t}^{\text{weather}},$$

where $\tilde{\alpha}_{i,t}$ is a cross-sectionally shrunk factor-model intercept, $\tilde{f}_t$ is a forecast of factor risk premia, and $\text{MOM}_{i,t}$ is the industry’s 12-month cumulative return skipping the most recent

month. Factor premia are forecast using historical factor means and ridge regressions on lagged FRED macro predictors.

Economic Weather Overlay

The distinguishing feature is an Economic Weather overlay. It classifies the macro environment using only lagged information, then enters the optimizer through expected returns, position limits, and risk appetite.

Weather states are determined as follows. A storm occurs when the term spread is inverted or unemployment is rising. A heatwave occurs when inflation is above its rolling median and short rates are rising. Sunshine occurs when industrial production growth is strong and unemployment is not rising. Other months are classified as goldilocks.

Weather state	Macro condition	Return tilt	Construction effect
Sunshine	Strong industrial production and stable labor market	Cyclical engines, technology networks, household beta	Lower γ_t and higher caps for favored groups
Goldilocks	No extreme macro pressure	Networks, household beta, selected cyclicals	Slightly lower γ_t with moderate active risk
Heatwave	High inflation and rising short rates	Extractors, utilities, inflation-sensitive industries	Slightly higher γ_t and higher caps for extractors
Storm	Inverted curve or rising unemployment	Defensive shelter industries	Higher γ_t and tighter caps for cyclicals and financials

The return overlay $\omega_{i,t}^{weather}$ is an economic prior: storms favor defensive industries; sunshine favors cyclicals and technology; heatwaves favor extractors and utilities; goldilocks favors software, hardware, chips, labs, and business services.

The covariance matrix is estimated with the same rolling factor model:

$$\hat{\Sigma}_t = \hat{B}_t \hat{\Sigma}_{f,t} \hat{B}_t' + \hat{D}_t,$$

where $\hat{B}_t$ contains industry factor betas and $\hat{D}_t$ contains residual variances. A small shrinkage term toward a diagonal covariance target improves numerical stability and reduces overfit.

Given $\hat{\mu}_t$ and $\hat{\Sigma}_t$, the strategy solves a benchmark-relative active mean-variance problem against the equal-weight industry benchmark b :

$$\max_w \quad w' \hat{\mu}_t - \frac{\gamma_t}{2} (w - b)' \hat{\Sigma}_t (w - b).$$

subject to

$$\sum_i w_i = 1, \quad 0 \leq w_i \leq c_{i,t}.$$

The cap $c_{i,t}$ is regime dependent: favored industries receive a 24% cap, neutral industries 20%, and disfavored industries 14%. Storms give defensive shelter industries more room while restricting cyclicals and balance-sheet-sensitive industries; heatwaves favor extractors; sunshine favors cyclicals and technology.

The optimizer searches over risk-aversion parameters and selects the best combination of ex-ante Sharpe ratio and active information ratio. The grid is scaled into γ_t by weather: sunshine uses 0.90, goldilocks 0.97, heatwave 1.03, and storm 1.15. Thus the optimizer takes slightly more active risk in supportive environments and becomes more conservative when recession risk rises.

Backtesting Results

The simulated out-of-sample period runs from January 1990 through October 2025, ending with the last month available after aligning the AQR and FRED data with the industry returns. Each month, the strategy forms weights using the prior 120 months and then holds the portfolio for one month. The following table reports annualized performance statistics based on monthly excess returns.

Portfolio	Ann. excess return	Ann. volatility	Sharpe	Max drawdown
49-industry optimized strategy	11.75%	15.20%	0.77	-43.53%
Equal-weight 49 industries	9.06%	16.04%	0.57	-54.01%
Market excess return	8.93%	15.24%	0.59	-54.11%
10-industry robustness strategy	10.56%	14.36%	0.74	-39.81%

The optimized 49-industry strategy earns an annualized excess return of 11.75% with annualized volatility of 15.20%, producing a Sharpe ratio of 0.77. This improves on both the equal-weighted industry benchmark and the market excess return, which have Sharpe ratios of 0.57 and 0.59, respectively, over the same period. The weather overlay, regime-dependent caps, and regime-dependent risk aversion also reduce the strategy's maximum drawdown relative to the non-weather specification.

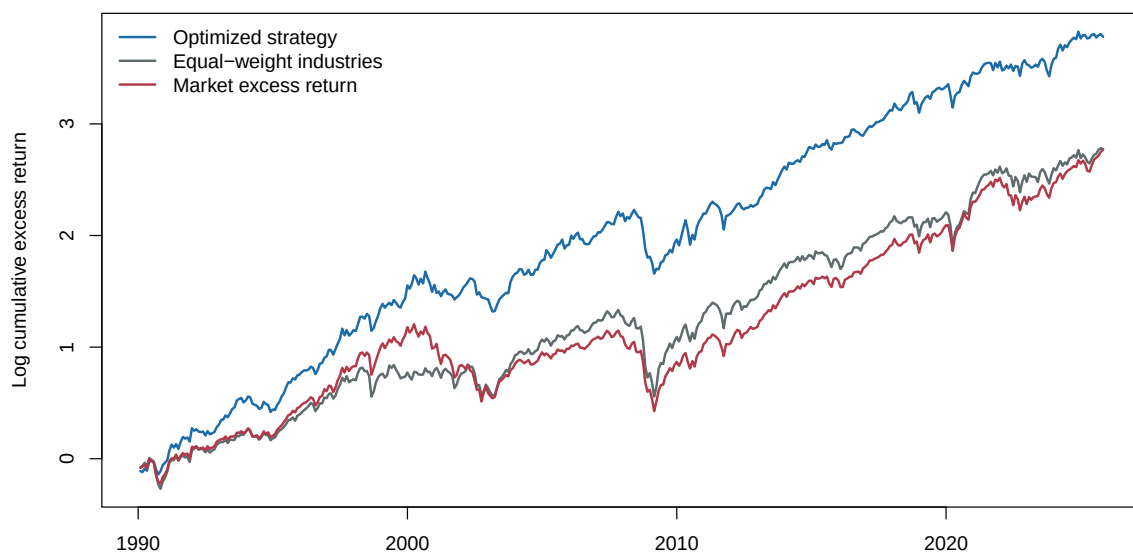


Figure 1: Log cumulative excess returns for the optimized strategy and benchmarks.

The cumulative return plot shows that the strategy's advantage is not driven by a single short episode. It tends to compound faster than the benchmarks after major drawdowns, while still sharing broad equity-market risk.

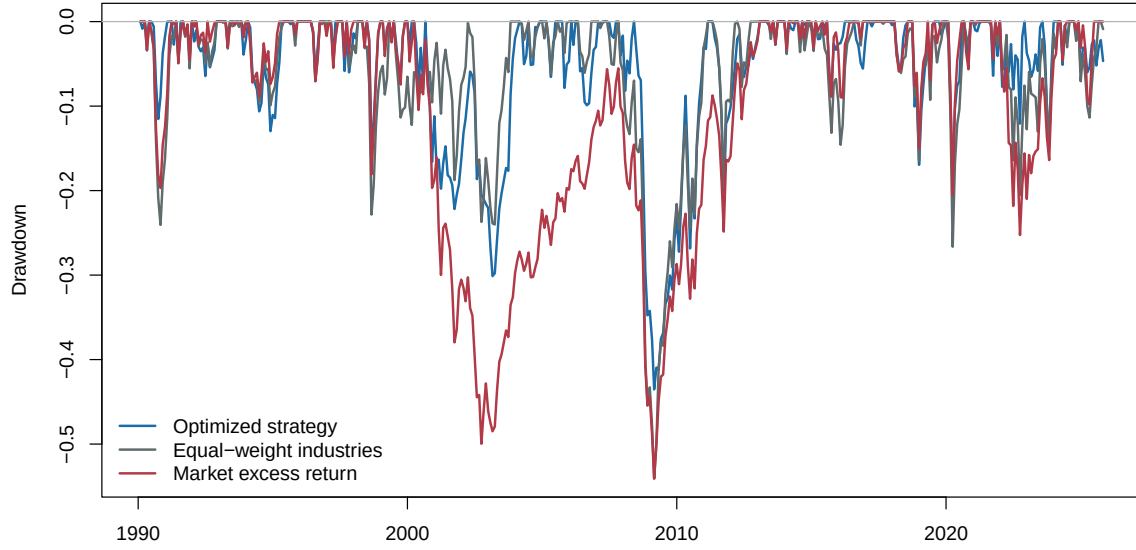


Figure 2: Drawdowns based on log cumulative excess returns.

The weather classification is not merely cosmetic. The strategy performs best in sunshine and heatwave environments, while storm months remain the hardest regime. This is economically sensible because storms include inverted curves and rising unemployment, when equity risk is broadly difficult to time.

Weather	Months	Ann. excess return	Ann. volatility	Sharpe	Ann. active return
Goldilocks	83	11.23%	15.41%	0.73	4.83%
Heatwave	69	12.88%	14.16%	0.91	1.24%
Storm	151	7.23%	16.49%	0.44	-0.92%
Sunshine	127	16.85%	14.00%	1.20	6.36%

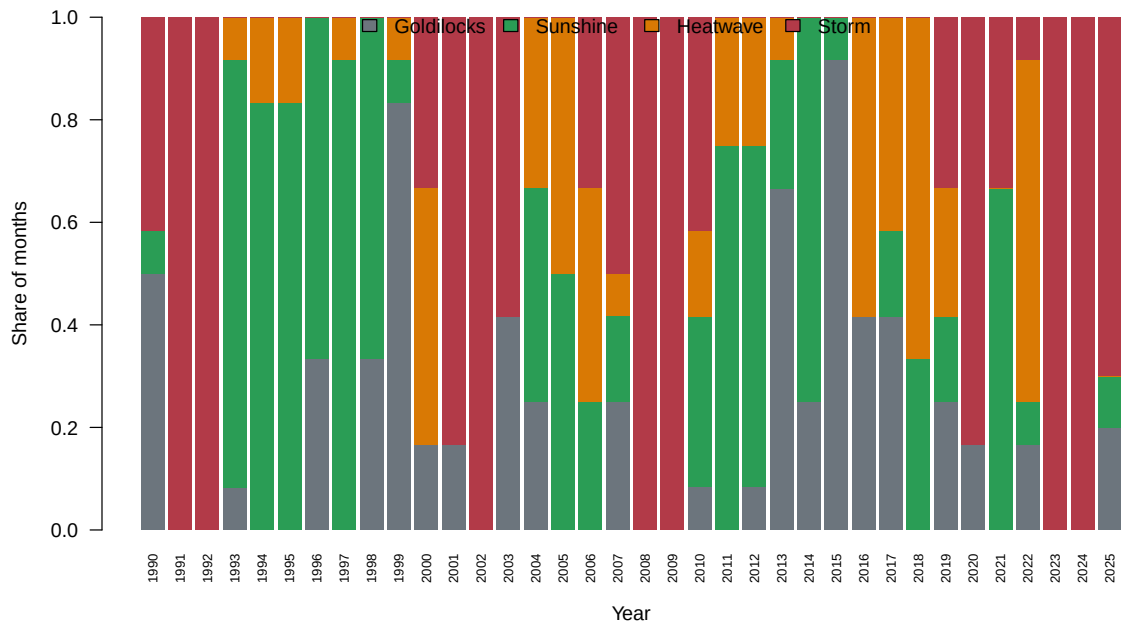


Figure 3: Annual composition of economic weather regimes.

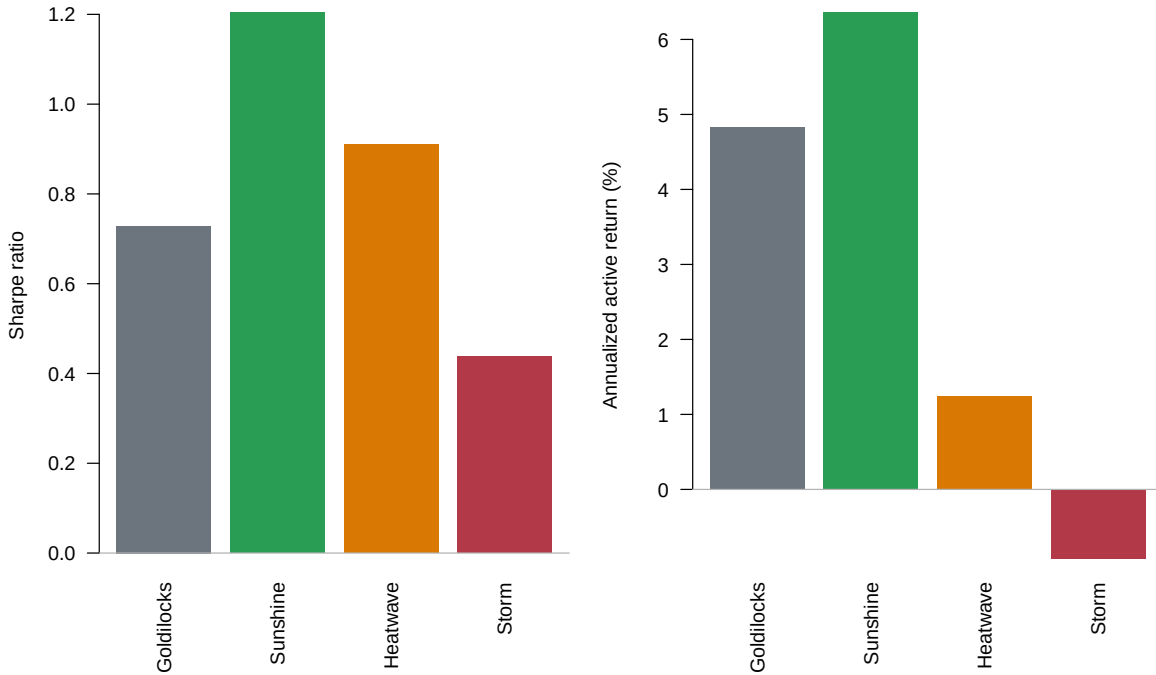


Figure 4: Strategy performance by economic weather regime.

Factor-adjusted performance is more modest. The strategy’s annualized Fama-French six-factor alpha is slightly negative and statistically insignificant, while its market beta is close to one. This suggests that the strategy’s advantage is better interpreted as improved timing and allocation across industry, macro, quality, and defensive exposures rather than a large standalone anomaly unexplained by common factors.

Portfolio	Monthly alpha	Annualized alpha	Alpha t-stat	Market beta	Adj. R2
49-industry optimized strategy	-0.02%	-0.18%	-0.17	0.98	0.84
Equal-weight 49 industries	-0.10%	-1.22%	-2.02	1.01	0.95

Because the strategy is designed relative to the equal-weight industry benchmark, its active performance is also important. The annualized active return is positive, although the information ratio is moderate rather than spectacular.

Portfolio	Ann. active return	Tracking error	Information ratio
vs equal-weight 49 industries	2.69%	9.08%	0.30

The full augmented-factor regression also clarifies what the portfolio is holding. The strongest positive exposures are to market, momentum, profitability, investment, and BAB; QMJ is negative over this sample, so the improvement should not be interpreted as a pure quality premium.

Term	Loading	t-stat
Alpha	0.01%	0.07
Mkt-RF	0.95	37.03
SMB	0.11	3.48
HML	0.04	1.03
RMW	0.27	4.60
CMA	0.11	2.02
Mom	0.35	16.11
QMJ	-0.16	-2.63
BAB	0.07	2.43
Adj. R2	0.84	

Turnover is economically relevant because the strategy rebalances monthly. The average monthly one-way turnover is shown below.

Metric	Value
Average monthly one-way turnover	27.86%
Median monthly one-way turnover	26.47%

Transaction costs reduce performance but do not eliminate the strategy's risk-adjusted advantage. Even with 25 bps of one-way trading cost applied to monthly turnover, the net Sharpe ratio remains above 0.72.

One-way cost	Ann. excess return	Ann. volatility	Sharpe	Max drawdown
0 bps	11.75%	15.20%	0.77	-43.53%
5 bps	11.58%	15.20%	0.76	-43.59%
10 bps	11.42%	15.20%	0.75	-43.64%
25 bps	10.92%	15.20%	0.72	-43.81%

The allocation results show that the optimizer does not simply buy one or two industries throughout the sample. Average weights are spread across defensive, consumer, technology, and retail industries, while the rolling weights reveal meaningful time variation.

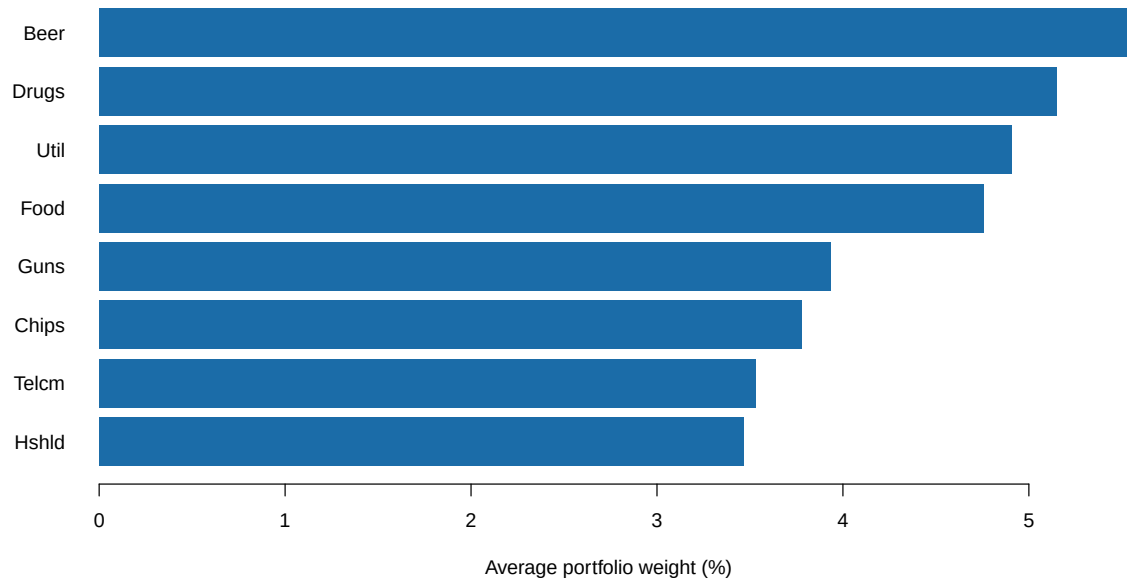


Figure 5: Top average industry weights over the out-of-sample period.

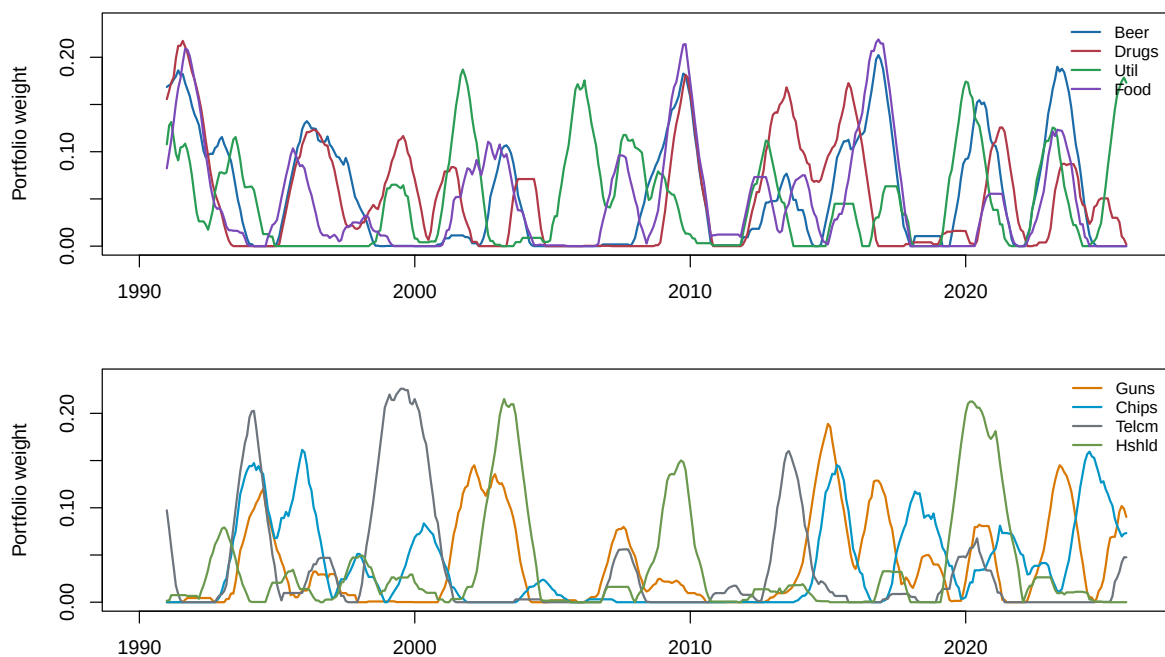


Figure 6: Rolling 12-month average weights, split across the top eight average allocations.

The strategy is not uniformly strong in every decade. It performs especially well in the 1990s and 2010s, is roughly flat relative to the equal-weight benchmark during the 2000s, and lags somewhat during the short 2020-2025 subsample. This is a useful limitation: the model improves long-run risk-adjusted performance but does not remove regime dependence.

Subpe- riod	Strategy ann. excess return	Strategy Sharpe	Equal-weight Sharpe	Market Sharpe
1990- 1999	16.65%	1.16	0.63	0.93
2000- 2009	5.41%	0.34	0.27	-0.10
2010- 2019	14.82%	1.03	0.86	1.01
2020- 2025	8.97%	0.54	0.59	0.74

The robustness test using 10 broader industries produces a Sharpe ratio of 0.73, close to the main result. This is encouraging because it indicates that the basic idea is not entirely dependent on the finer 49-industry classification. The 10-industry strategy also has lower

volatility and a smaller maximum drawdown, although its return is less distinctive because broad industries reduce the amount of cross-sectional opportunity.

Conclusion

The evidence supports a cautious conclusion: an economic-weather industry-rotation strategy can improve risk-adjusted performance relative to simple equity benchmarks. The strongest result is the improvement in Sharpe ratio and drawdown relative to both the equal-weight industry portfolio and the market. The weaker result is that the factor alpha is not statistically significant, so the strategy should be interpreted as disciplined allocation across economic habitats rather than a large standalone anomaly.

The strategy is feasible to implement with liquid industry ETFs or sector portfolios, and the transaction-cost sensitivity suggests that moderate trading costs would not overturn the main result. Future improvements could include Bayesian shrinkage for factor-premium forecasts or direct constraints on factor exposures. These extensions may improve implementation realism while staying within the same portfolio optimization framework.

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